

Proof. Let $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n)$ be vectors of real numbers. Define the inner product and the associated Euclidean norms by

$$\langle a, b \rangle := \sum_{i=1}^n a_i b_i, \quad \|a\|^2 := \langle a, a \rangle = \sum_{i=1}^n a_i^2, \quad \|b\|^2 := \langle b, b \rangle = \sum_{i=1}^n b_i^2.$$

Zero-vector case. If $\|b\|^2 = 0$ then each $b_i = 0$. Hence

$$\langle a, b \rangle = \sum_{i=1}^n a_i \cdot 0 = 0, \quad \|a\|^2 \|b\|^2 = \|a\|^2 \cdot 0 = 0,$$

so that $\langle a, b \rangle^2 = 0 = \|a\|^2 \|b\|^2$. The same argument works when $\|a\|^2 = 0$. Consequently we may assume from now on that $\|a\|^2 > 0$ and $\|b\|^2 > 0$.

A non-negative quadratic. For a real parameter t set

$$Q(t) := \|a - tb\|^2 = \sum_{i=1}^n (a_i - tb_i)^2.$$

Each term $(a_i - tb_i)^2$ is a square, therefore

$$Q(t) \geq 0 \quad \text{for all } t \in \mathbb{R}. \quad (1)$$

Expanding and summing yields

$$\begin{aligned} Q(t) &= \sum_{i=1}^n (a_i^2 - 2ta_i b_i + t^2 b_i^2) \\ &= \sum_{i=1}^n a_i^2 - 2t \sum_{i=1}^n a_i b_i + t^2 \sum_{i=1}^n b_i^2 \\ &= \|a\|^2 - 2t \langle a, b \rangle + t^2 \|b\|^2. \end{aligned}$$

Thus $Q(t) = At^2 + Bt + C$ with

$$A = \|b\|^2 \geq 0, \quad B = -2\langle a, b \rangle, \quad C = \|a\|^2 > 0.$$

Discriminant argument. A quadratic polynomial $At^2 + Bt + C$ that satisfies $A \geq 0$ and $At^2 + Bt + C \geq 0$ for every real t must have a non-positive discriminant:

$$\Delta := B^2 - 4AC \leq 0.$$

Applying this to the polynomial $Q(t)$ (which fulfills the hypothesis by (1)) gives

$$\begin{aligned}
0 &\geq B^2 - 4AC \\
&= (-2\langle a, b \rangle)^2 - 4\|b\|^2\|a\|^2 \\
&= 4\langle a, b \rangle^2 - 4\|a\|^2\|b\|^2.
\end{aligned}$$

Dividing by the positive constant 4 we obtain the Cauchy–Schwarz inequality

$$\boxed{\langle a, b \rangle^2 \leq \|a\|^2 \|b\|^2}. \quad (3)$$

Equality case. Equality in the discriminant condition occurs exactly when $\Delta = 0$; then the quadratic has a double root. From (2) we can write

$$Q(t) = \|b\|^2(t - t_0)^2, \quad t_0 = \frac{\langle a, b \rangle}{\|b\|^2}.$$

Hence $Q(t_0) = 0$ and, by the definition of Q ,

$$0 = Q(t_0) = \|a - t_0 b\|^2 = \sum_{i=1}^n (a_i - t_0 b_i)^2.$$

A sum of non-negative squares is zero only if each square vanishes, i.e.

$$a_i = t_0 b_i \quad \text{for all } i = 1, \dots, n.$$

Thus equality in (3) holds iff there exists a scalar λ (namely $\lambda = t_0$) such that $a_i = \lambda b_i$ for every i . Conversely, if $a_i = \lambda b_i$ for some $\lambda \in \mathbb{R}$, then

$$\langle a, b \rangle = \lambda \sum_{i=1}^n b_i^2 = \lambda \|b\|^2, \quad \|a\|^2 = \lambda^2 \|b\|^2,$$

and a direct substitution shows $\langle a, b \rangle^2 = \|a\|^2 \|b\|^2$. The case $\lambda = 0$ corresponds to one of the vectors being the zero vector and is included in the statement.

Conclusion. For arbitrary real sequences $\{a_i\}_{i=1}^n$ and $\{b_i\}_{i=1}^n$ we have

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right),$$

with equality exactly when the two vectors are proportional (or one of them is the zero vector). $\square$